



Winning Space Race with Data Science

Fallon Robinson-Wynn
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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
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- References

Executive Summary

- Summary of methodologies
 - Five methods were utilized to complete the thorough data analysis of the SpaceX Falcon 9 dataset. Four different machine learning models were developed, trained, evaluated, and compared.
- Summary of all results
 - The optimal predictive model that best estimates the likelihood of successful Falcon 9 first-stage landings is the KNN Classification model.
 - The KSC LC-39A launch site has the highest success rate for Falcon 9 first-stage landings.
 - The FT & B4 Booster Versions are optimal for successful landings.
 - The GTO, LEO, ISS, & SSO Orbit Types are optimal for successful landings.
 - The Ground Pad & Drone Ship landing types are optimal for successful landings.

Introduction

Project background and context:

To reduce launch costs and gain a competitive advantage, this project analyzes historical SpaceX Falcon 9 first-stage landing data to build a predictive model that estimates the likelihood of successful landings, which is a key factor in determining future launch cost efficiency.

Problems you want to find answers:

- Rate of successful first-stage launches.
- Factors that influence first-stage successful landings.
- Launch site with best success rate
- Cost efficiency of Future rocket launches

Section 1

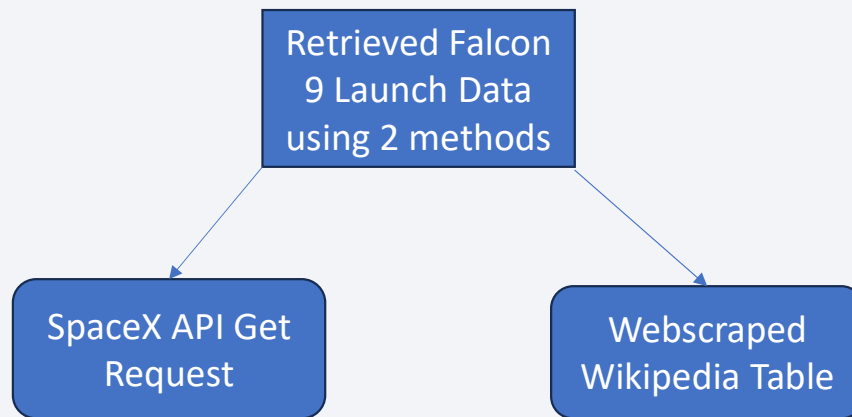
Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Describe how data was collected using API & Web Scraping
- Performed data wrangling
 - Describe how data was processed using Python applications
- Performed exploratory data analysis (EDA) using visualization and SQL
- Performed interactive visual analytics using Folium and Plotly Dash
- Performed predictive analysis using classification models
 - How to build, tune, evaluate classification models

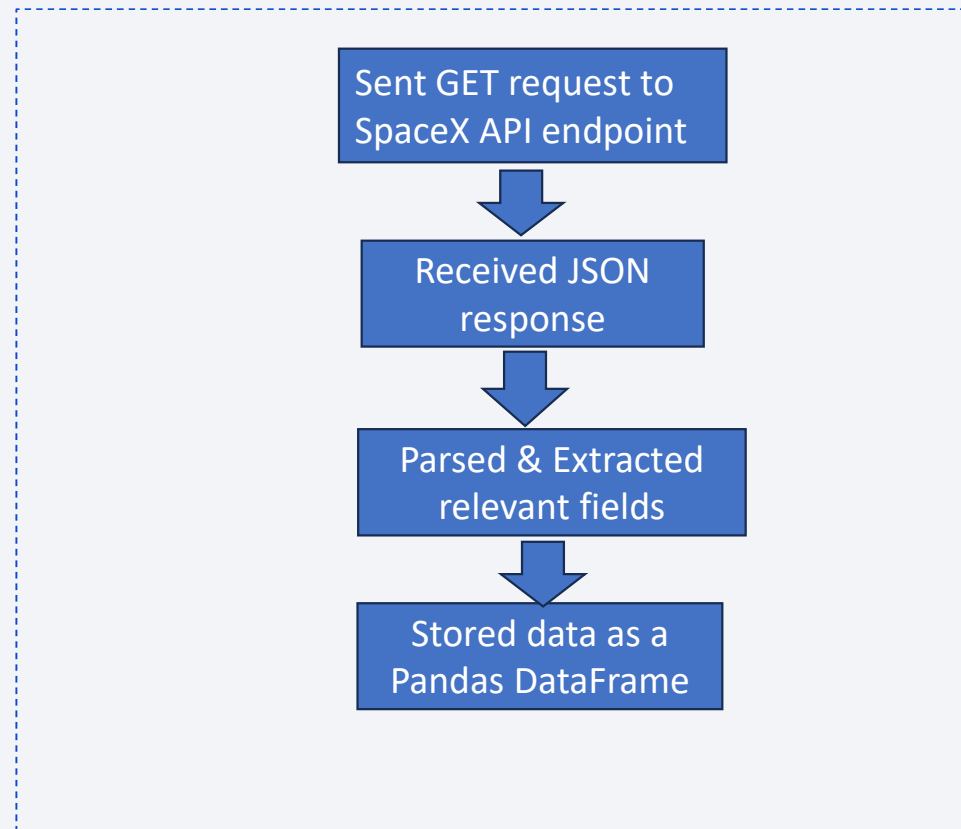
Data Collection



Data Collection – SpaceX API (Application Programming Interface)

GitHub URL of the completed
SpaceX API calls notebook:

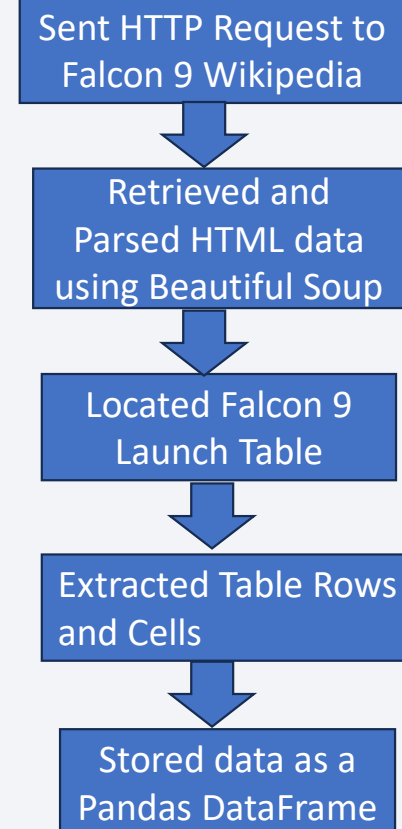
<https://github.com/fallonwynn/TestRepo/blob/main/SpaceX%20API%20Data%20Collection.ipynb>



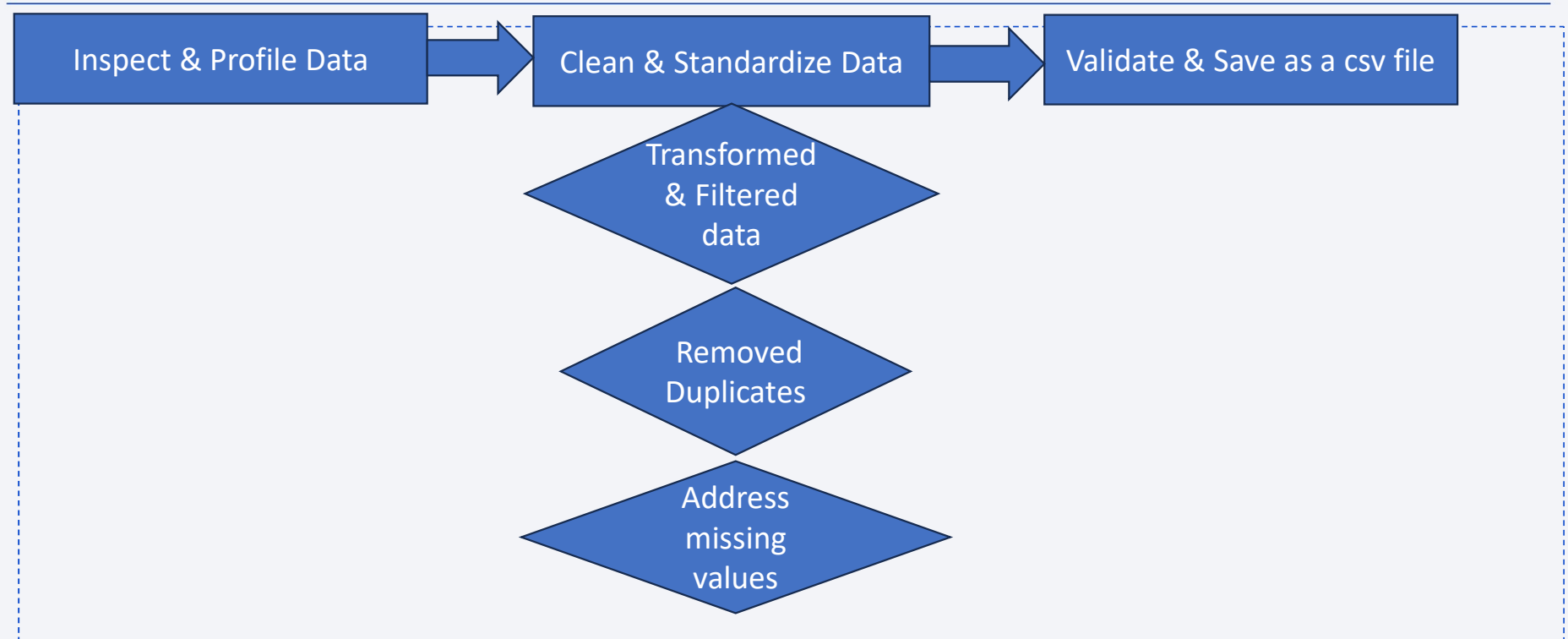
Data Collection – Web Scraping

GitHub URL of the completed web scraping notebook:

<https://github.com/fallonwynn/TestRepo/blob/main/SpaceX-webscraping.ipynb>



Data Wrangling



GitHub URL of the completed data wrangling notebooks:

<https://github.com/fallonwynn/TestRepo/blob/main/SpaceX-Data%20wrangling.ipynb>

<https://github.com/fallonwynn/TestRepo/blob/main/SpaceX%20API%20Data%20Collection.ipynb>

API Falcon 9 Data Features

- **Date:** Day the rocket was launched
- **FlightNumber:** Sequential number of the Falcon 9 launch
- **BoosterVersion:** Specific model or version of the Falcon 9 booster used
- **PayloadMass:** Weight (Kg) of the payload carried into orbit
- **Orbit:** Type of orbit the payload was sent to
- **LaunchSite:** Physical location where the rocket was launched
- **Outcome:** If the mission was successful or unsuccessful
- **Flights:** Number of times that specific booster has flown
- **GridFins:** Did the booster have grid fins installed for landing control
- **Reused:** If the booster reused from a previous mission
- **Legs:** Were landing legs attached for recovery
- **LandingPad:** The landing location
- **Block:** The block number of the booster
- **ReusedCount:** The number of times booster previously used
- **Serial:** Unique booster id
- **Longitude:** longitude coordinate of the landing or launch location
- **Latitude:** latitude coordinate of the landing or launch location

Web Scrapping Falcon 9 Data Features

- **Flight No:** Sequential number of the Falcon 9 launch
- **Launch site:** Physical location where the rocket was launched
- **Payload:** The object or cargo the rocket is carrying
- **Payload mass:** Weight (Kg) of the payload carried into orbit
- **Orbit:** Type of orbit the payload was sent to
- **Customer:** Company performing the launch
- **Launch Outcome:** If the mission was successful or unsuccessful
- **Version Booster:** Specific model or version of the Falcon 9 booster used
- **Booster landing:** The landing location
- **Date:** Day the rocket was launched
- **Time:** Time the rocket was launched

Data Cleaning & Standardization

- Removed rows with single rocket that had multiple cores and payloads
- Check datatypes of each feature.
- Converted the date_utc to a datetime datatype.
- Filtered dataframe to include only the dates of launches for 11/13/20 and prior.
- Filtered dataframe to include only Falcon 9 launches.
- Filled missing PayloadMass values with the mean PayloadMass value.
- The LandingPad column has missing values that represents when they were not used and was not changed.
- Created new feature “Class” to categorize all successful outcomes as “1” and unsuccessful outcomes as “0”.

EDA with Data Visualization

- Scatter Plot to visualize the relationship between Flight Number & Launch Site.
- Scatter Plot to visualize the relationship between Payload Mass & Launch Site.
- Bar Chart to visualize the success rate of each orbit type.
- Scatter Plot to visualize the relationship between Flight Number & Orbit type.
- Scatter Plot to visualize the relationship between Payload Mass & Orbit type.
- Line Plot to visualize the launch success yearly trend.

GitHub URL of the completed EDA with data visualization notebook:

<https://github.com/fallonwynn/TestRepo/blob/main/SpaceX%20Data%20Visualization.ipynb>

EDA with SQL

- Displayed distinct launch sites
- Displayed 5 records of launch sites beginning with the string 'CCA'.
- Displayed NASA (CRS) total payload mass
- Displayed booster version F9 v1.1 average payload mass
- Displayed date of first successful ground pad landing
- Displayed booster version with both success in drone ship and payload mass between 4000 and 6000

GitHub URL of the completed data wrangling notebooks:

<https://github.com/fallonwynn/TestRepo/blob/main/Exploratory%20Data%20Analysis%20of%20SpaceX%20Data.ipynb>

EDA with SQL Continued

- Displayed total number of successful and unsuccessful missions
- Used a subquery to list all the booster versions that have carried the maximum payload mass
- Displayed the month, landing outcome, booster version, and launch site for launches that occurred in 2015
- Displayed the count of 2 landing outcomes between 2010-06-04 and 2017-03-20 in descending order

GitHub URL of the completed data wrangling notebooks:

<https://github.com/fallonwynn/TestRepo/blob/main/Exploratory%20Data%20Analysis%20of%20SpaceX%20Data.ipynb>

Built an Interactive Map with Folium

- Added a circle marker and a popup name label for each launch site to easily visualize the physical location on a Folium map.
- Added colored marker clusters to easily visualize launch outcomes at each site, where green=successful and red=unsuccessful.
- Added distance line and distance label to mark the proximities of launch site to the coastline, nearest railroad, nearest city, and nearest highway.

GitHub URL of the completed interactive map with Folium map:

https://github.com/fallonwynn/TestRepo/blob/main/SpaceX_launch_site_location_Folium.ipynb

Built a Dashboard with Plotly Dash

Created the SpaceX Launch Records Dashboard

Pie Chart

- Added to easily visualize the total successful launches by site.
- Can select from a drop-down menu to select “All Sites” or a specific site.
- Selecting one site displays the ratio of successful/unsuccessful landings for that site

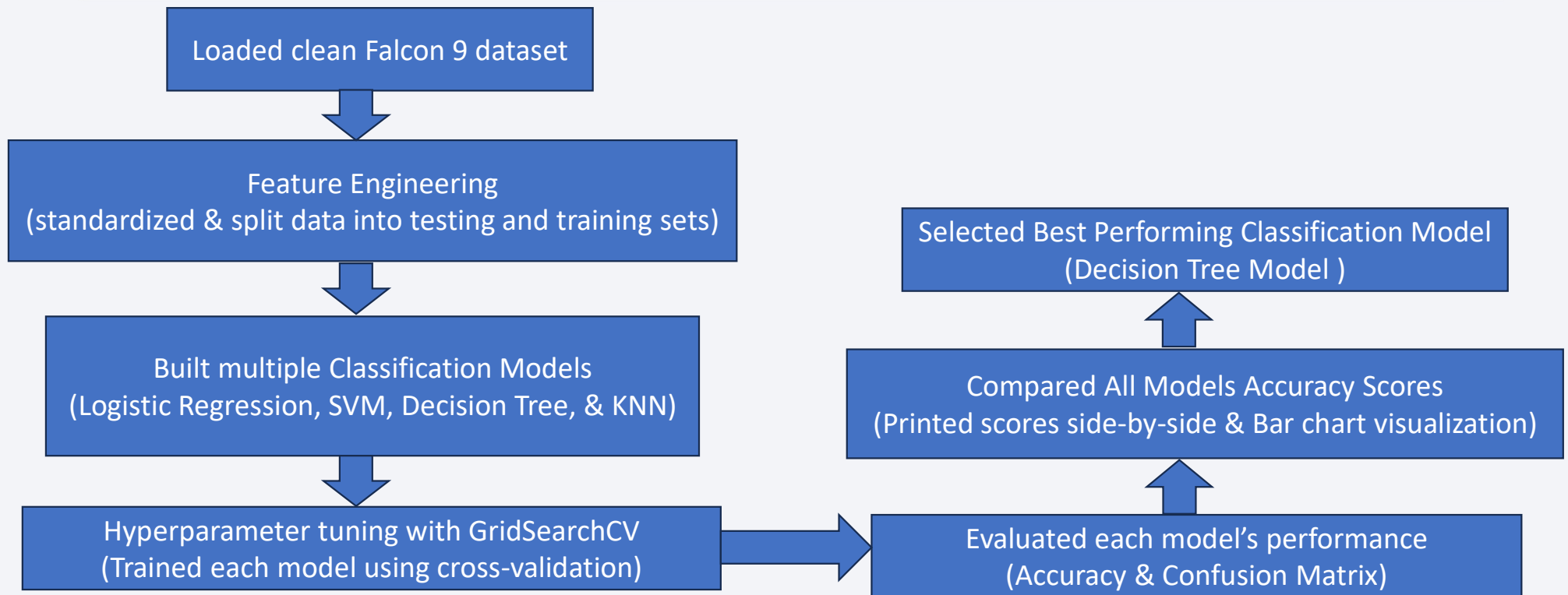
Scatter Plot

- Added to easily visualize how payload mass affects the success rate of different booster versions.
- Added a range slider for Payload Mass for drilling down based on different ranges.

GitHub URL of the completed Plotly Dash lab:

<https://github.com/fallonwynn/TestRepo/blob/main/spacex-dash-app.py>

Predictive Analysis (Classification)



GitHub URL of the completed predictive analysis lab:

19

https://github.com/fallonwynn/TestRepo/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

Exploratory data analysis results

- CCAFS SLC-40/ CCAFS LC-40 had the most total launches
- CCAFS SLC-40/ CCAFS LC-40 rockets mainly carried cargo between 0 and 8,000 kg
- KSC LC-39A had the second most total launches
- KSC LC 39A's rockets mainly carried cargo between 2,000 and 7,000 kg
- Orbits ES-L1, GEO, HEO, & SSO have equally 100% success rate across range of payloads
- ES-L1, GEO, HEO, & SO were used only once
- The SSO orbit is one of the newest used and was used only 5 times
- VLEO orbit has a success rate of 90% across range of payloads.
- The VLEO orbit is the third newest used and used 14 times.
- The LEO orbit was the 1ST to be used and has 100% successful launches with payloads between 1,000 and 5,500 kg
- The GTO orbit was the 4TH to be used and has a success rate of 80% with payloads between 3,000 & 5,300 kg
- ISS orbit was the 2ND to be used and has a success rate of 60% with payloads between 0 & 5,500 kg
- The highest success rates occurred between 2015 & 2020

Results Continued

Interactive analytics demo in screenshots

- The KSC LC-39A launch site has the highest success rate
 - 76.9% of launches were successful
 - 23.1% of launches failed
 - The most successful outcomes was with FT Booster version with payloads between 0 and 5,500 kg, which were the most used version.
 - The second most successful outcomes was with B4 Booster version with payloads between 3,000 and 5,000 kg
- Overall, Rockets with version FT boosters have the highest launch success rate with payload ranges from 475-5,300 and overall is the version with the highest success rate.
- Overall, Rockets with version B4 boosters have the highest launch success rate with payload ranges from 3,310-3,700 and overall is the version with the 2nd highest success rate.
- Booster version F9 FT B1022, F9 FT B1026, F9 FT B1021.2, & F9 FT B1031.2 that had payloads between 4000 & 6000 have successfully landed on drone ship.

Results Continued

KSC LC-39A Falcon 9 Equipment STATS

Booster Version	Orbit Type	Successful Landing Type
FT: 75%	GTO: 67% ISS: 22% LEO: 11%	Ground Pad: 50% Drone Ship: 50%
B4: 25%	GTO: 33.3% ISS: 33.3% LEO: 33.3%	Ground Pad: 66.7% Drone Ship: 33.3%
Combined Usage	GTO: 50.2% ISS: 27.6% LEO: 22.2%	Ground Pad: 55.6% Drone Ship: 44.4%

Results Continued

Predictive analysis results

Classification Model	Best Parameters	Training Accuracy	Testing Accuracy	Impression
Logistic Regression	C': 0.01, 'penalty': 'l2', 'solver': 'lbfgs'	84.6%	83.3%	Performs Well
Decision Tree	criterion': 'gini', 'max_depth': 16, 'max_features': 'sqrt', 'min_samples_leaf': 1, 'min_samples_split': 10, 'splitter': 'random'	87.7%	66.7%	Overfitted
SVM	'C': 1.0, 'gamma': 0.03162277660168379, 'kernel': 'sigmoid'	84.82142857142856%	83.3%	Performs Well
KNN	'algorithm': 'auto', 'n_neighbors': 10, 'p': 1	84.82142857142858%	83.3%	Best Performing

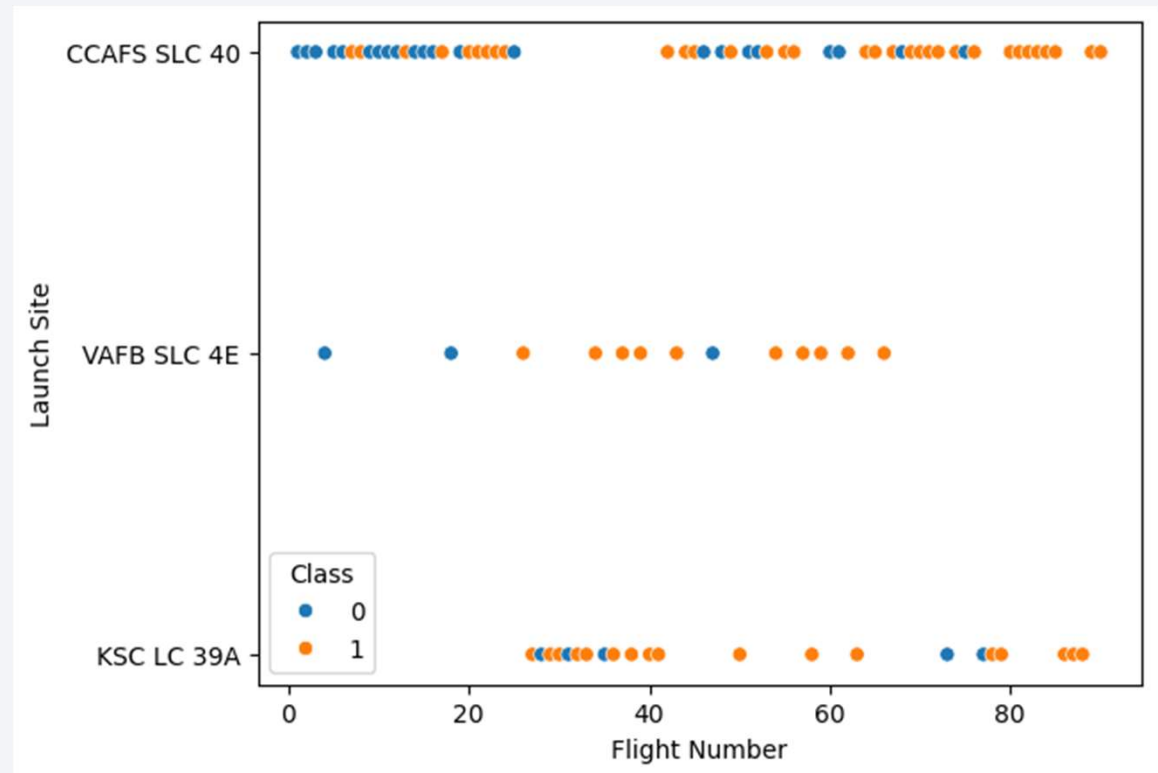
The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a dynamic pattern of diagonal streaks in shades of blue, red, and cyan on the right. These streaks are layered over a fine, light-colored grid, creating a sense of depth and movement, reminiscent of a digital or data visualization theme.

Section 2

Insights drawn from EDA

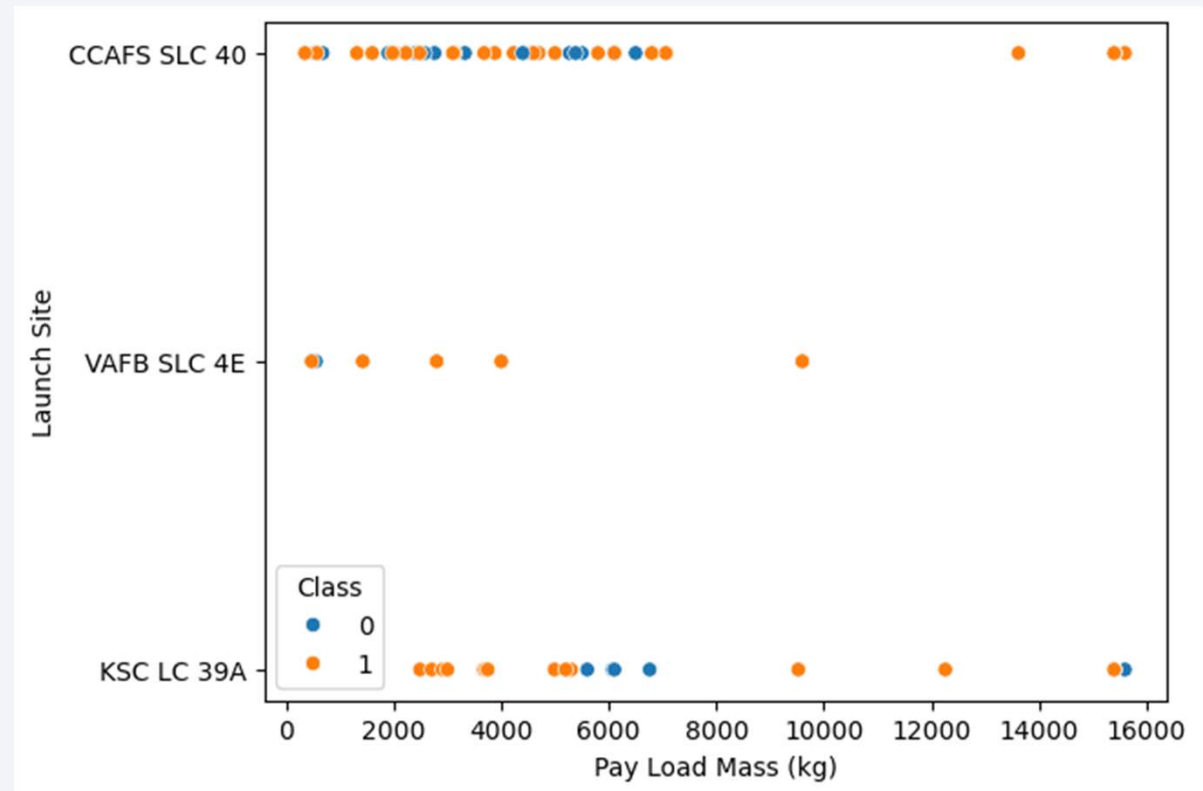
Flight Number vs. Launch Site

- CCAFS SLC 40 had the most total launches
- KSC LC-39A had the second most total launches
- VAFB SLC 4E had the least total launches.
- Over time, all sites had more consistent successful launches.



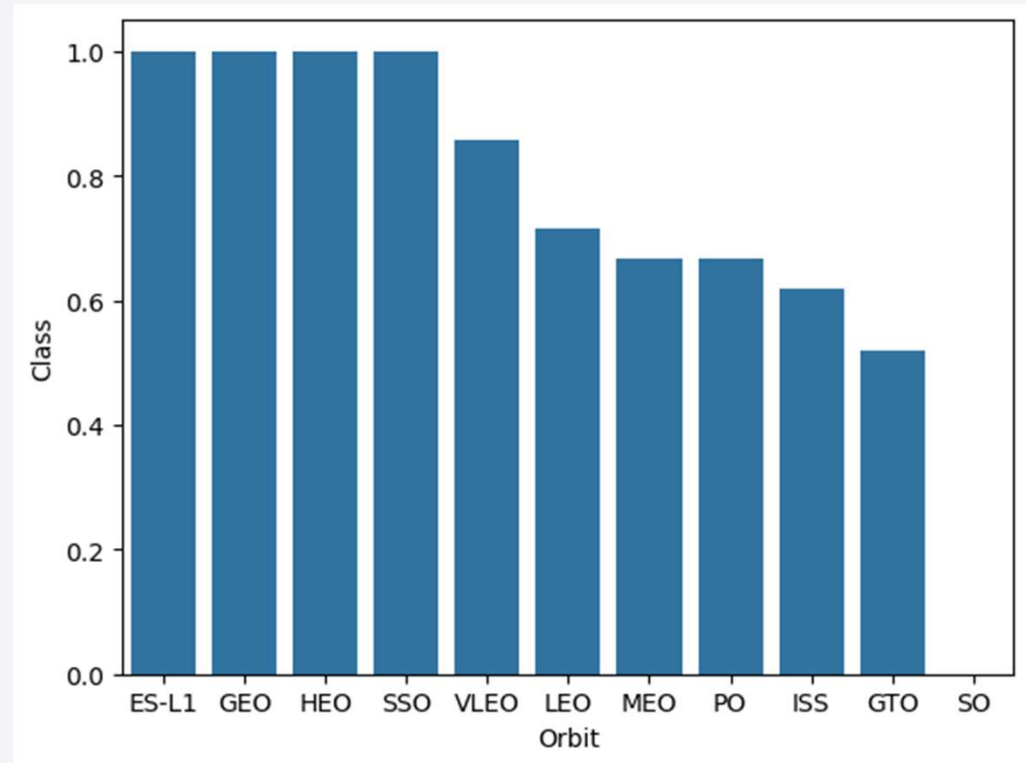
Payload vs. Launch Site

- CCAFS SLC 40 had the most successful launches with payload masses $>10,000$ kg then any other site
- CCAFS SLC 40 rockets mainly carried cargo between 0 and 8,000 kg
- VAFB SLC 4E rockets mainly carried cargo between 0 and 5,000 kg
- KSC LC 39A rockets mainly carried cargo between 2,000 and 7,000 kg



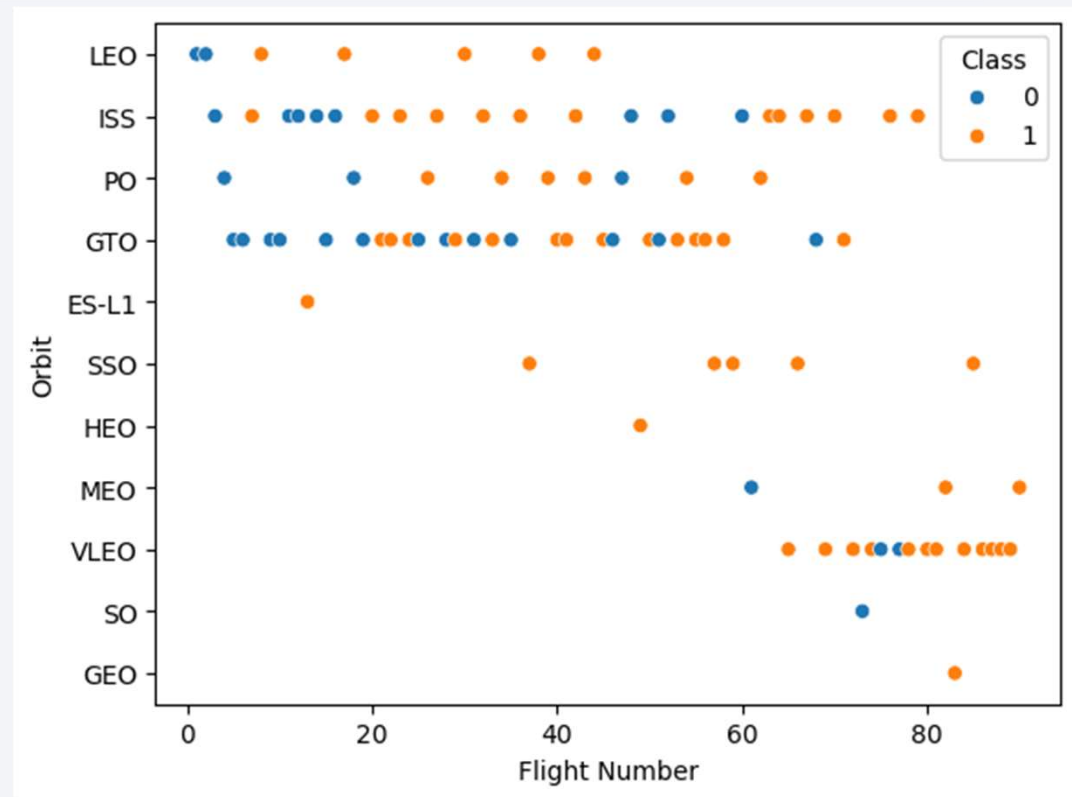
Success Rate vs. Orbit Type

- Orbits ES-L1, GEO, HEO, & SSO have equally 100% success rate
- VLEO orbit has a success rate of approximately 90%.
- LEO orbit has a success rate of approximately 70%
- Orbits MEO & PO have equally 65% success rate
- ISS orbit has a success rate of approximately 62%
- GTO orbit has a success rate of 50%
- SO orbit has the lowest success rate of 0%



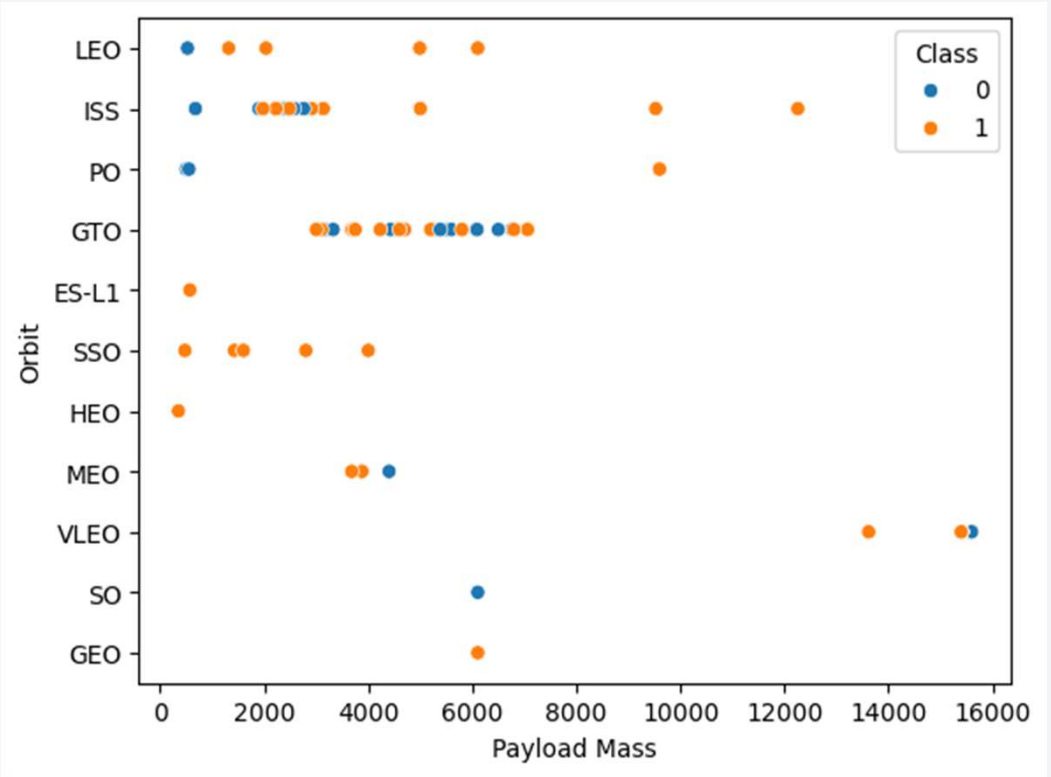
Flight Number vs. Orbit Type

- The LEO orbit is the 1ST to be used.
- The ISS orbit is the 2ND to be used
- Orbits PO & GTO are the 3RD & 4TH to be used, respectively
- The GEO orbit is the newest and used only once.
- ES-L1, HEO, & SO were used only once
- The MEO orbit was used only 3 times
- The SSO orbit was used only 5 times
- The VLEO orbit is the third newest used and used 14 times.



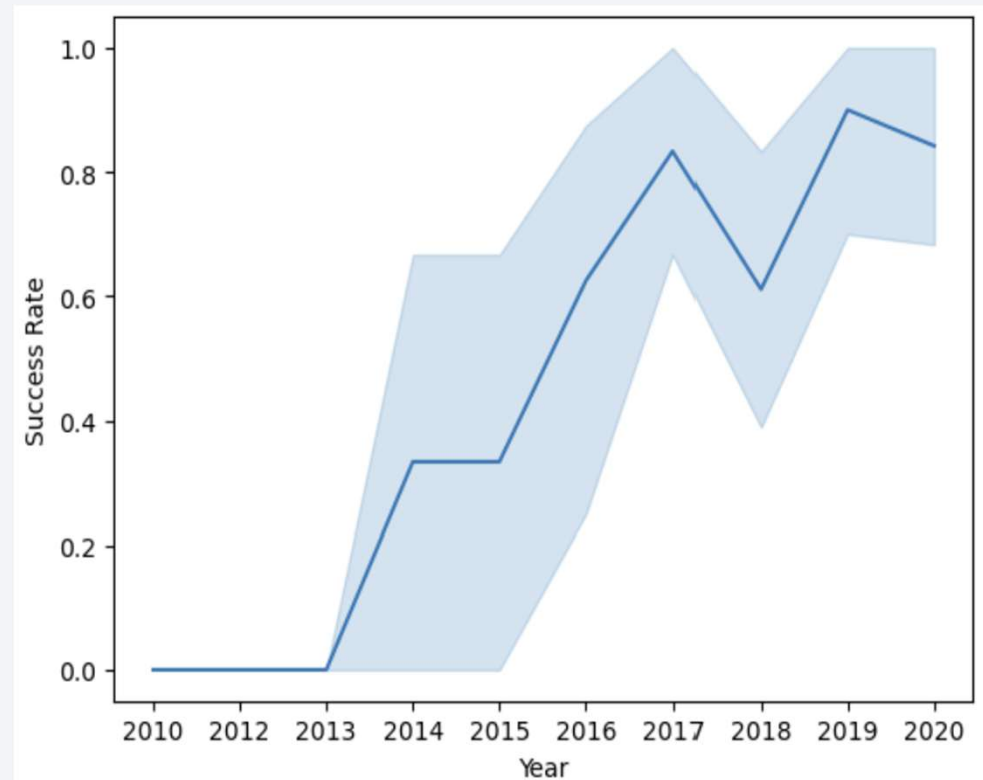
Payload vs. Orbit Type

- The VLEO orbit is associated with payloads $> 13,000$
- ES-L1 & HEO orbits are associated payloads $< 1,000$
- ISS, PO, & VLEO have successful launches at higher payloads
- SSO has all successful launches with payloads ranging between 0 and 4,000
- GTO orbit payloads range between 3,000 and 8,000 with a close balance between outcomes



Launch Success Yearly Trend

- Between 2010 and 2013 all launches failed
- As of 2013, the number success rate increased until 2017.
- Between 2017 and 2018, the success rate decreased from approximately 85% to 60%
- Between 2018 and 2019, the success rate increased from 60% to 90%.
- Between 2019 and 2020, the success rate decreased slightly to approximately 87%.



All Launch Site Names

- SQL query used to find the names of the unique launch sites:
 - %sql SELECT Distinct("Launch_Site") FROM SPACEXTBL;

Launch Site Abbreviation	Launch Site Full Name	Location
CCAFS LC-40	Cape Canaveral Space Force Station – Launch Complex 40	Florida
VAFB SLC-4E	Vandenberg Space Force Base – Space Launch Complex 4 East	California
KSC LC-39A	Kennedy Space Center – Launch Complex 39A	Florida
CCAFS SLC-40	Cape Canaveral Space Force Station – Space Launch Complex 40	Florida

- There were 4 distinct launch sites displayed, however CCAFS LC-40 & CCAFS SLC-40 are technically the same location.

Launch Site Names Begin with 'CCA'

- SQL query used to find 5 records where launch sites begin with 'CCA':

```
%sql SELECT * From SPACEXTBL WHERE "Launch_Site" LIKE "CCA%" LIMIT 5;
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

The table displays the first 5 early launches from Cape Canaveral Air Force Station (CCAFS LC-40)

Total Payload Mass

- SQL query used to Calculate the total payload carried by boosters from NASA:

```
%sql SELECT SUM("PAYLOAD_MASS__KG_") AS "NASA(CRS) Total Payload Mass" FROM  
SPACEXTBL WHERE Customer = 'NASA (CRS)';
```

NASA(CRS) Total Payload Mass
45596

- The query shows that across all missions NASA(CRS) carried 45,596 kg of cargo to space.

Average Payload Mass by F9 v1.1

- SQL query used to calculate the average payload mass carried by booster version F9 v1.1:

```
%sql SELECT ROUND(AVG("PAYLOAD_MASS__KG_"), 2) AS "F9 v1.1 Average Payload  
Mass" FROM SPACEXTBL WHERE Booster_Version LIKE '%F9 v1.1%';
```

F9 v1.1 Average Payload Mass

2534.67

- The query shows that booster version F9 v1.1 typically carried an average of 2,535 kg of cargo to space.

First Successful Ground Landing Date

- SQL query used to find the dates of the first successful landing outcome on ground pad:

```
%sql SELECT MIN(DATE) AS "First Successful Ground Pad Landing" FROM SPACEXTBL  
WHERE Landing_Outcome = "Success (ground pad)" LIMIT 1;
```

First Successful Ground Pad Landing

2015-12-22

- The query shows that the first successful ground pad landing occurred on December 22, 2015.

Successful Drone Ship Landing with Payload between 4000 and 6000

- Query used to list the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000:

```
%sql SELECT DISTINCT("Booster_Version") FROM SPACEXTBL WHERE  
"Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS__KG_" BETWEEN  
4000 AND 6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- The query shows 4 distinct boosters

Total Number of Successful and Failure Mission Outcomes

- SQL query used to calculate the total number of successful and failure mission outcomes:

```
%sql SELECT CASE WHEN "Mission_Outcome" LIKE 'Success%' THEN 'Success' ELSE  
Failure'END AS Outcome_Type, COUNT(*) AS Total FROM SPACEXTBL GROUP BY  
Outcome_Type;
```

Outcome_Type	Total
Failure	1
Success	100

- The query shows there were 100 successful missions and 1 failed mission.

Boosters Carried Maximum Payload

- SQL query used to list the names of the booster which have carried the maximum payload mass:

```
%sql SELECT "Booster_Version" FROM SPACEXTBL WHERE "PAYLOAD_MASS__KG_"=
(SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTBL);
```

- The query shows 12 booster versions

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- SQL query used to list the failed landing_outcomes in drone ship, their booster versions, and launch site names for year 2015:

```
%sql SELECT substr(Date, 6,2) AS "Month", "Landing_Outcome", "Booster_Version",  
"Launch_Site" FROM SPACEXTBL WHERE substr(Date, 0,5)='2015' AND  
"Landing_Outcome" ='Failure (drone ship)';
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- The query shows that there were 2 failed landing outcomes in drone ship in 2015.

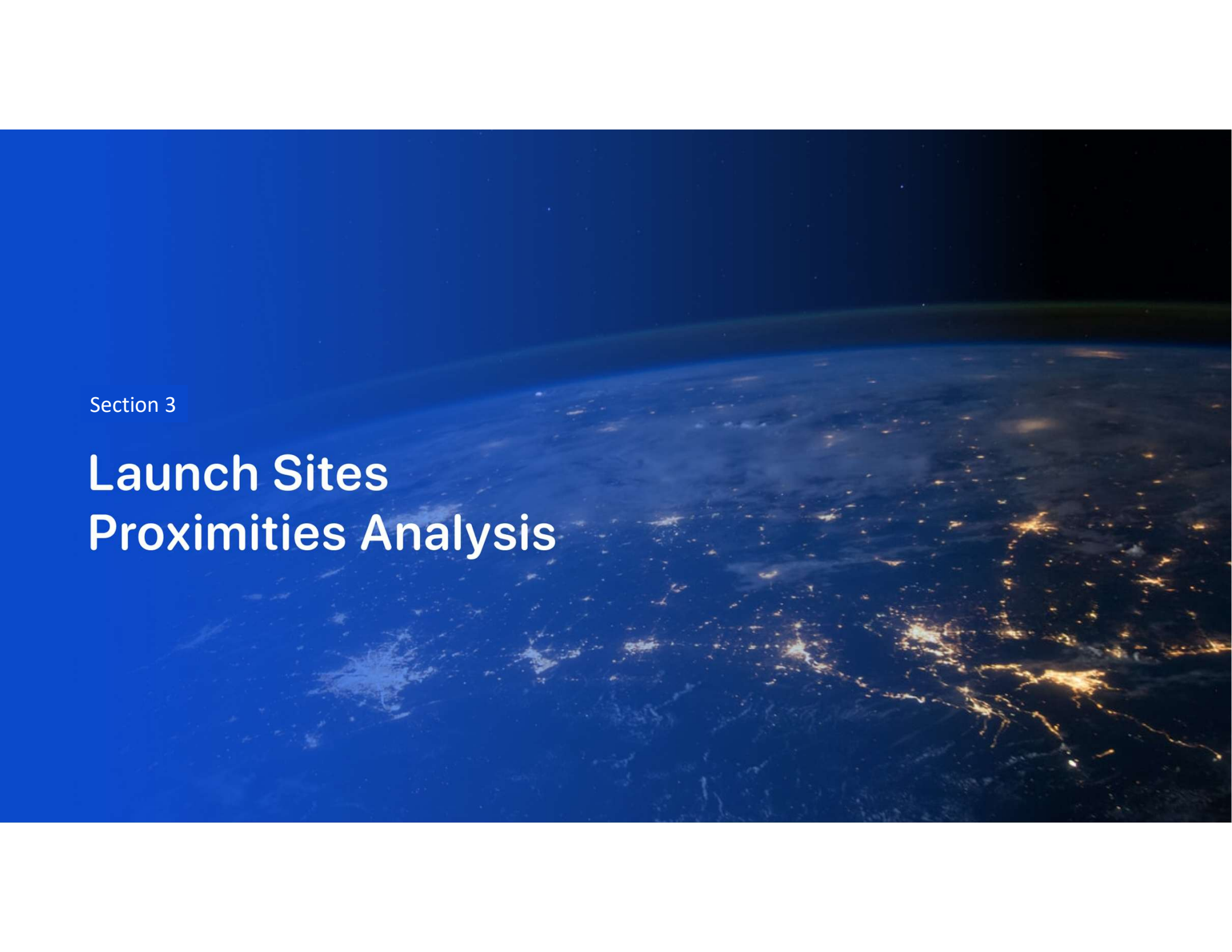
Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- SQL query used to rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order:

```
%sql SELECT "Landing_Outcome", COUNT("Landing_Outcome") AS "Total Count" FROM  
SPACEXTBL WHERE "Landing_Outcome" IN ('Failure (drone ship)', 'Success (ground pad)')  
AND DATE BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY "Landing_Outcome"  
ORDER BY "Total Count" DESC;
```

Landing_Outcome	Total Count
Failure (drone ship)	5
Success (ground pad)	3

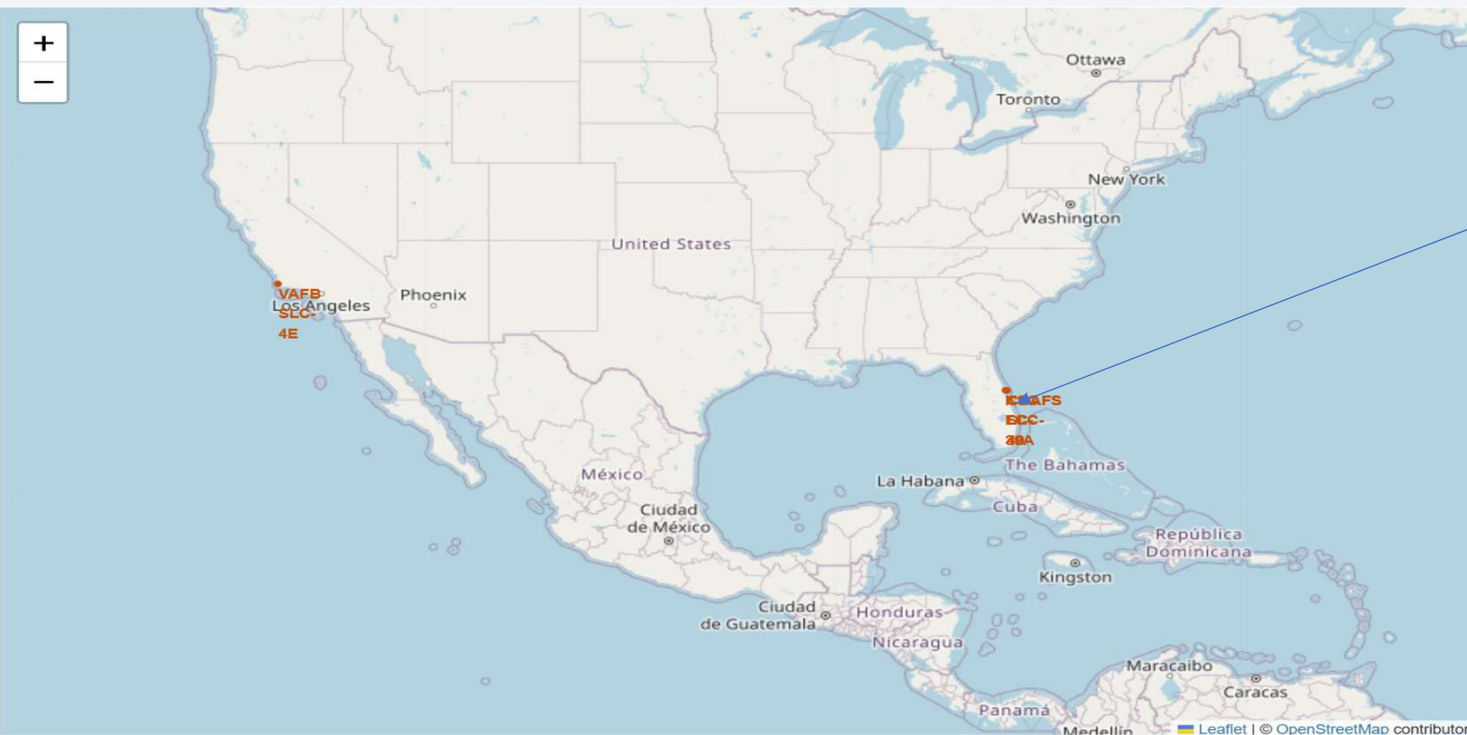
- The query shows that there were 5 drone ship failures and 3 ground pad successful landings between 6/4/10 & 3/20/17

A satellite view of Earth from space, showing the curvature of the planet and the glow of city lights at night. The image is used as a background for the title slide.

Section 3

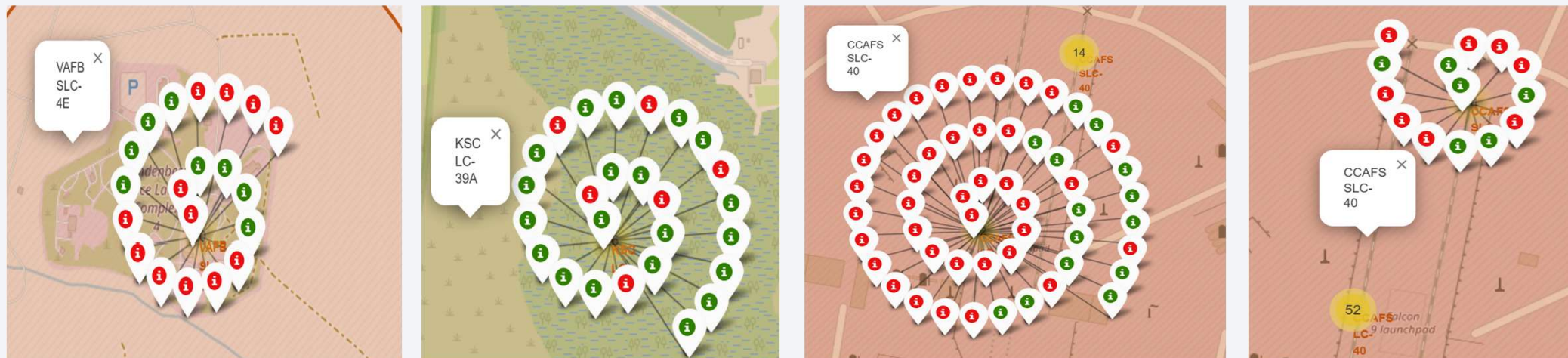
Launch Sites Proximities Analysis

Launch Sites Marked on a Folium Map



- The map shows VAFB SLC-4E is located off the coast of California
- The map shows that CCAFS SLC-40, CCAFS LC-40, & KSC LC-39A are located off the coast of Florida
- The map shows that CCAFS SLC-40 & CCAFS LC-40 are overlapping.

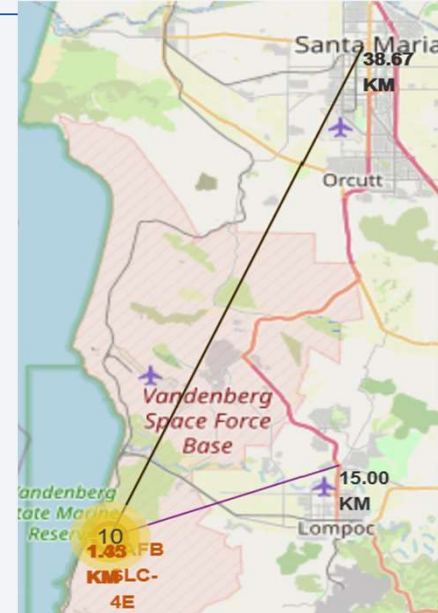
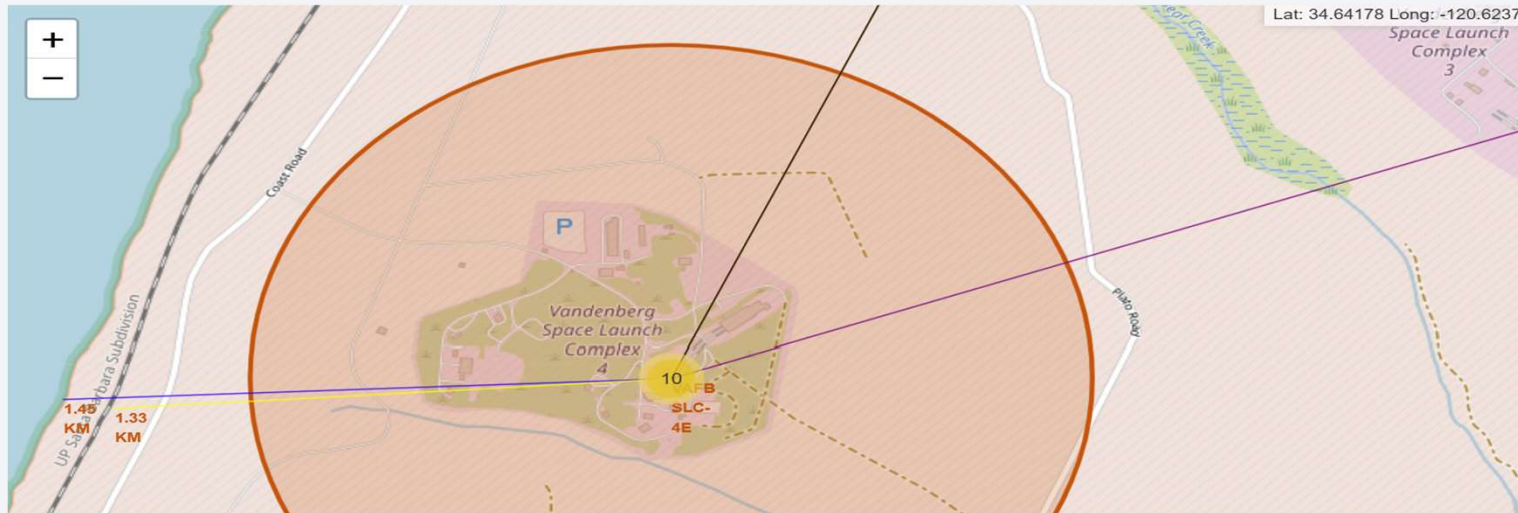
Launch Site Outcomes Displayed on Folium Map



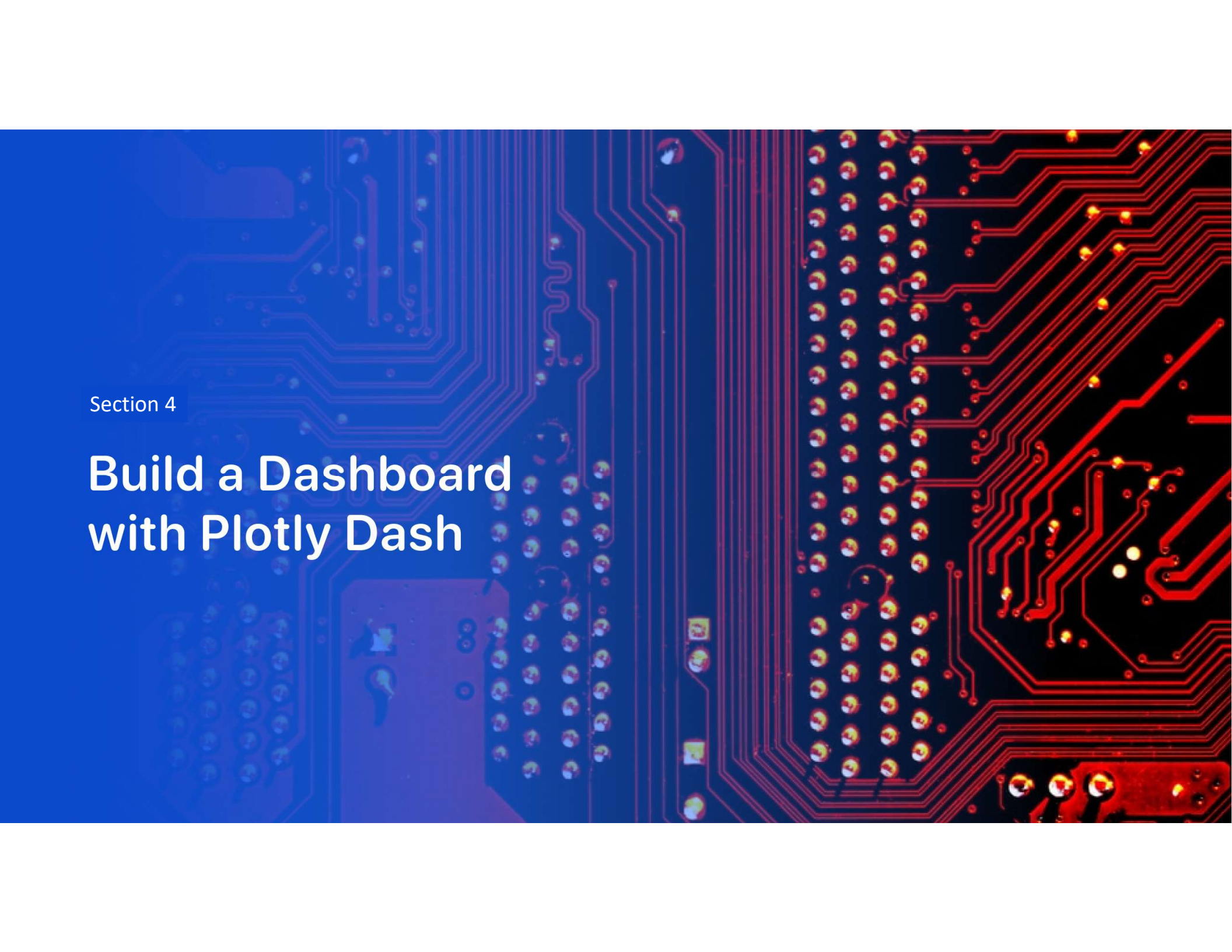
Note: Folium Map = success rate per site.

- KSC LC-39A has the most successful launches
- VAFB SLC-4E has the second most successful launches
- CCAFS SLC-40 has the least amount of successful launches
- Overall, CCAFS SLC-40 had the greatest number of launches, while VAFB SLC-40 had the lowest number of launches

Launch Site Proximities Marked on Folium Map



- VAFB SLC-4E is 1.45 Km from California's coastline and 1.33 Km from a railroad.
- VAFB SLC-4E is 38.67 Km from Santa Maria and 15.00 Km from the nearest highway (Cabrillo Highway).



Section 4

Build a Dashboard with Plotly Dash

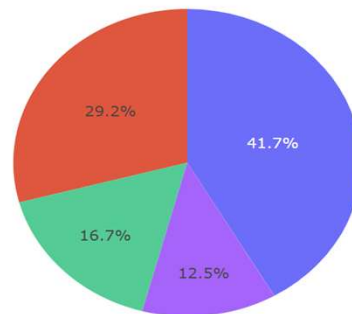
SpaceX Launch Outcomes Interactive Dashboard

SpaceX Launch Records Dashboard

All Sites



Total Successful Launches by Site



■ KSC LC-39A
■ CCAFS LC-40
■ VAFB SLC-4E
■ CCAFS SLC-40

Note: Dashboard Pie Chart = percentage of total successes contributed by each site.

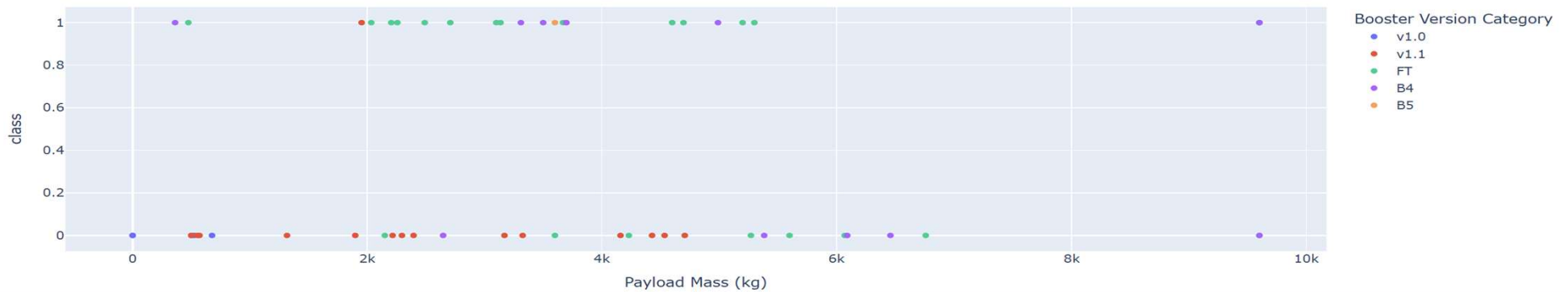
- KSC LC-39A has the highest success rate.
- CCAFS LC-40 has the second highest success rate
- VAFB SLC-4E has the third highest success rate
- CCAFS SLC-40 has the lowest success rate

All Sites-Payload vs Outcome Interactive Dashboard

Payload range (Kg):



Payload vs. Outcome for All Sites



- Rockets with version FT boosters have the highest launch success rate with payload ranges from 475-5,300 and overall is the version with the highest success rate.
- Rockets with version B4 boosters have the highest launch success rate with payload ranges from 3,310-3,700 and overall is the version with the 2nd highest success rate.
- Payload mass does not appear to influence the success rate of Rockets with version v1.1 boosters since the vast majority of outcomes were failures over the complete range from 500-4,707.
- Rockets with version v1.1 boosters have the 2nd lowest launch success rate.
- There were only 2 outcomes for rockets with version v1.0, one payload was 0 and the other was 677, but both were failures having the lowest launch success rate at 0%.
- There was only 1 outcome for one rocket with version B5 booster, which was successful at payload 3,600.

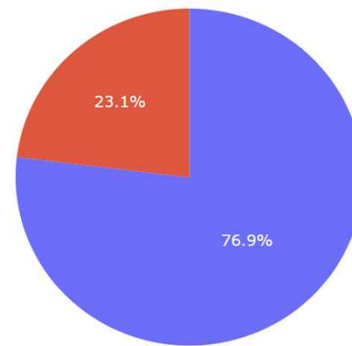
KSC LC-39A Has the Highest Success Rate

SpaceX Launch Records Dashboard

KSC LC-39A

× ▾

Successful vs. Failed Launches at KSC LC-39A



■ 1
■ 0

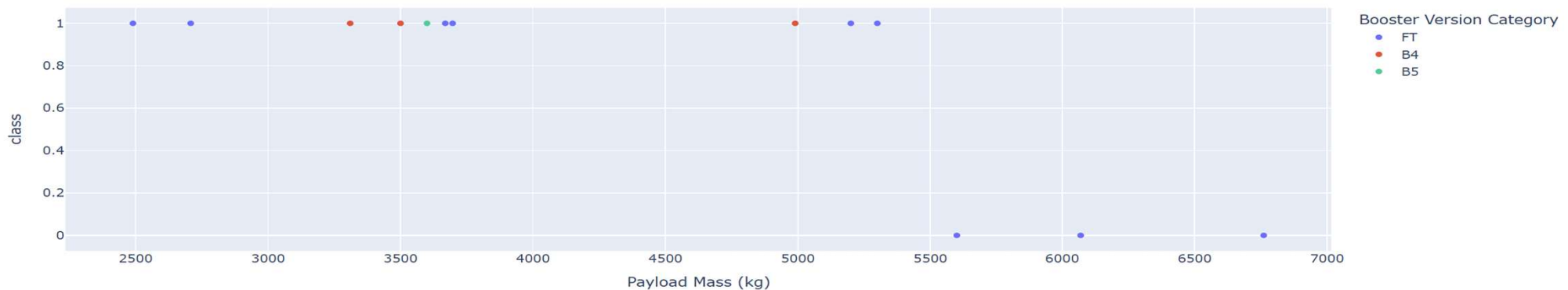
- 76.9% of launches were successful
- 23.1% of launches failed

KSC LC-39A Payload vs Outcome

Payload range (Kg):



Payload vs. Outcome for KSC LC-39A



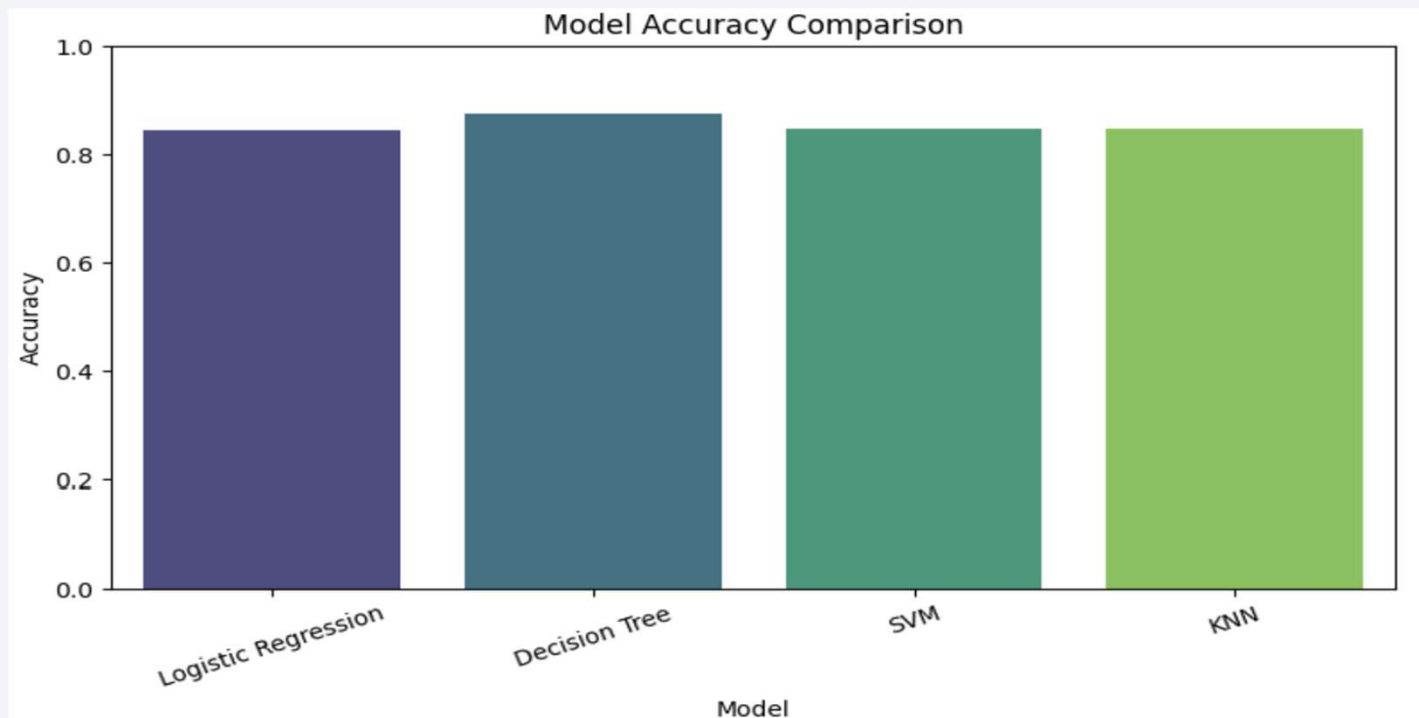
- The most successful outcomes was with FT Booster version with payloads between 0 and 5,500 kg, which were most used version.
- The second most successful outcomes was with B4 Booster version with payloads between 3,000 and 5,000 kg.
- B5 Booster version was only used once and was successful with the payload of approximately 3,600 kg.
- FT Booster version had unsuccessful outcomes with payload between 5,500 and 7,000.

The background of the slide features a dynamic, abstract image. On the left, there is a solid blue area. To the right, a tunnel-like structure is depicted with curved, flowing lines in shades of blue and white, creating a sense of motion and depth. The lines curve around a central point, leading the eye towards the right side of the frame.

Section 5

Predictive Analysis (Classification)

Classification Accuracy



The Decision Tree model has the highest classification accuracy at 87.7% on the training data

51

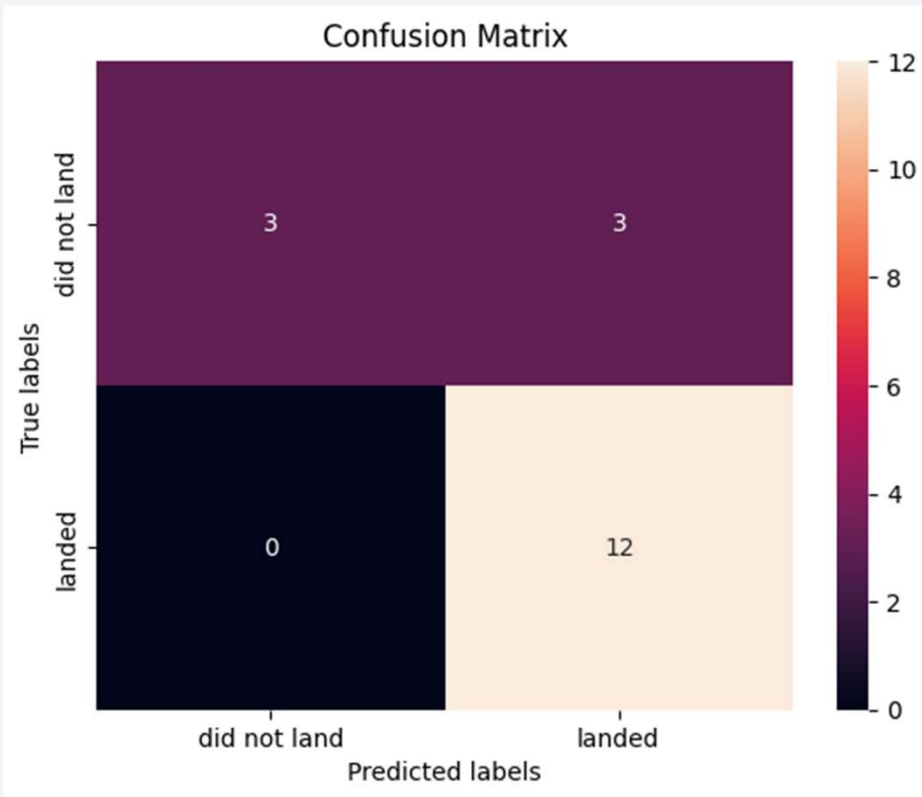
Note: See appendix to see data for All 4 models

Classification Accuracy

Model	Training Accuracy	Testing Accuracy	Impression
Logistic Regression	84.6%	83.3%	Performs well
SVM	84.82142857142856%	83.3%	Performs well
Decision Tree	87.7%	66.7%	Overfitted
KNN	84.82142857142858%	83.3%	Best Performing

- The Decision Tree model does not perform well on unseen data since the testing accuracy was lower at 66.7%.
- The best performing model is the KNN model since it had the 2nd highest training accuracy and a relatively close testing accuracy demonstrating slightly better generalization than the SVM model.

Confusion Matrix: KNN Classifier Model

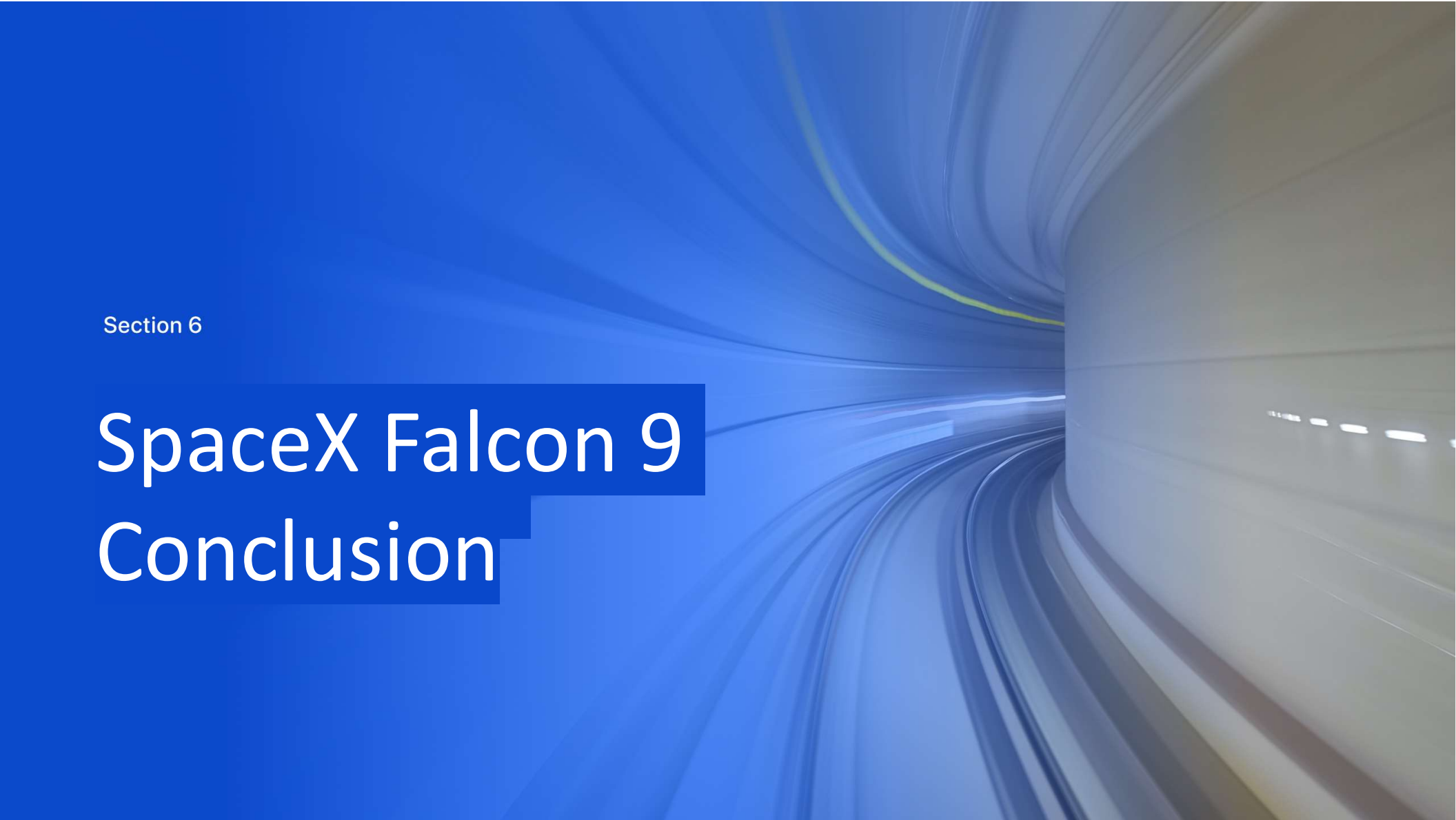


- 100% of first stage “did not land” outcomes were correctly predicted as unsuccessful landings.
- 0% of first stage “did not land” outcomes were incorrectly predicted as successful landings.
- 80% of first stage “landed” outcomes were correctly predicted as successful landings.
- 25% of first stage “landed” outcomes were incorrectly predicted as unsuccessful landings.
- The Logistic Regression & SVM models have the same confusion matrix results (see appendix).

Note: See appendix to see data for All 4 models

Section 6

SpaceX Falcon 9 Conclusion



Conclusions

- Since the KSC LC-39A launch site has the highest success rate for Falcon 9 first-stage landings the factors influencing their success should be modeled to achieve cost efficiency.
 - ❖ The following Booster Versions are optimal for successful landings:
 - ✓ FT Booster version with payloads between 0 and 5,500 kg
 - ✓ B4 Booster version with payloads between 3,000 and 5,000 kg
 - ❖ The following Orbit Types are optimal for successful landings:
 - ✓ LEO Orbit is the 1ST to be used and has 100% successful launches with payloads between 1,000 and 5,500 kg
 - ✓ GTO Orbit is then 4TH to be used and have 80% successful launches with payloads range between 3,000 and 5,500 kg
 - ✓ ISS Orbit is the 2ND to be used and has 60% successful launches with payloads between 0 and 5,500 kg
 - ✓ SSO Orbit has 100% successful launches with payloads ranging between 0 and 4,000 kg.
 - Although this orbit type was not used at KSC LC-39A and is one of the newer orbits, it is highly likely it would help increase the overall success rate.
 - ❖ The following landing types are optimal for successful landings:
 - ✓ Ground Pad (55.6% Successful)
 - ✓ Drone Ship (44.4% Successful)

Conclusions Continued

- Based on the available dataset, the KNN Classification model demonstrated the most stable performance and is therefore, the optimal predictive model that best estimates the likelihood of successful first-stage landings.
 - ✓ 84.8% of predicted Falcon 9 first-stage landings were accurate for the training data
 - ✓ 83.3% of predicted Falcon 9 first-stage landings were accurate for the testing data
 - ✓ Lowest variation across data sets.
- Data Limitations
 - ✓ There is limited data for the newer orbit types SSO, HEO, MEO, SO, & GEO.
 - ✓ There is limited data for the type of landing (Ground Pad, Drone Ship, Parachute, etc.)
 - ✓ Future analysis would provide a stronger analysis and more optimal performing predictive model with including additional data from 2021-2025.

Appendix

This function is to plot the confusion matrix.

```
def plot_confusion_matrix(y, y_predict):
    "this function plots the confusion matrix"
    from sklearn.metrics import confusion_matrix

    cm = confusion_matrix(y, y_predict)
    ax = plt.subplot()
    sns.heatmap(cm, annot=True, ax = ax); #annot=True to annotate cells
    ax.set_xlabel('Predicted labels')
    ax.set_ylabel('True labels')
    ax.set_title('Confusion Matrix');
    ax.xaxis.set_ticklabels(['did not land', 'landed']); ax.yaxis.set_ticklabels(['did not land', 'landed'])
    plt.show()
```

Load the dataframe

Load the data

```
from js import fetch
import io

URL1 = "https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/dataset_part_2.csv"
resp1 = await fetch(URL1)
text1 = io.BytesIO((await resp1.arrayBuffer()).to_py())
data = pd.read_csv(text1)
```

data.head()

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCc
0	1	2010-06-04	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	
4	5	2013-12-03	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	

Appendix

Load 2ND dataset and define X Features

```
URL2 = 'https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/dataset_part_3.csv'
resp2 = await fetch(URL2)
text2 = io.BytesIO((await resp2.arrayBuffer()).to_py())
X = pd.read_csv(text2)
```

```
X.head(100)
```

	FlightNumber	PayloadMass	Flights	Block	ReusedCount	Orbit_ES-L1	Orbit_GEO	Orbit_GTO	Orbit_HEO	Orbit_ISS	...	Serial_B1058	Serial_B10
0	1.0	6104.959412	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	(
1	2.0	525.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	(
2	3.0	677.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	1.0	...	0.0	(
3	4.0	500.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	(
4	5.0	3170.000000	1.0	1.0	0.0	0.0	0.0	1.0	0.0	0.0	...	0.0	(
...	...	...	...	...	...	...	...	...	...	...	...	...	(
85	86.0	15400.000000	2.0	5.0	2.0	0.0	0.0	0.0	0.0	0.0	...	0.0	(
86	87.0	15400.000000	3.0	5.0	2.0	0.0	0.0	0.0	0.0	0.0	...	1.0	(
87	88.0	15400.000000	6.0	5.0	5.0	0.0	0.0	0.0	0.0	0.0	...	0.0	(
88	89.0	15400.000000	3.0	5.0	2.0	0.0	0.0	0.0	0.0	0.0	...	0.0	(
89	90.0	3681.000000	1.0	5.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	(

90 rows × 83 columns

Appendix

Create numpy array for Y Class data

```
Y = data["Class"].to_numpy()
```

Standardize and Transform data

```
transform = preprocessing.StandardScaler()  
X_scaled = transform.fit_transform(X)
```

Split data into training and test sets

```
X_train, X_test, Y_train, Y_test = train_test_split(X_scaled, Y, test_size=0.2, random_state=2)
```

Appendix

1. Logistic Regression Model

- Created model

```
parameters = {"C": [0.01, 0.1, 1], 'penalty': ['l2'], 'solver': ['lbfgs']} # l1 lasso l2 ridge
lr = LogisticRegression()
logreg_cv = GridSearchCV(estimator=lr, cv=10, param_grid=parameters, scoring="accuracy", verbose=2)
logreg_cv.fit(X_train, Y_train)
```

Fitting 10 folds for each of 3 candidates. totalling 30 fits

- Tuned model & printed training accuracy & best parameters

```
print("tuned hyperparameters : (best parameters) ", logreg_cv.best_params_)
print("accuracy : ", logreg_cv.best_score_)

tuned hyperparameters : (best parameters) {'C': 0.01, 'penalty': 'l2', 'solver': 'lbfgs'}
accuracy : 0.8464285714285713
```

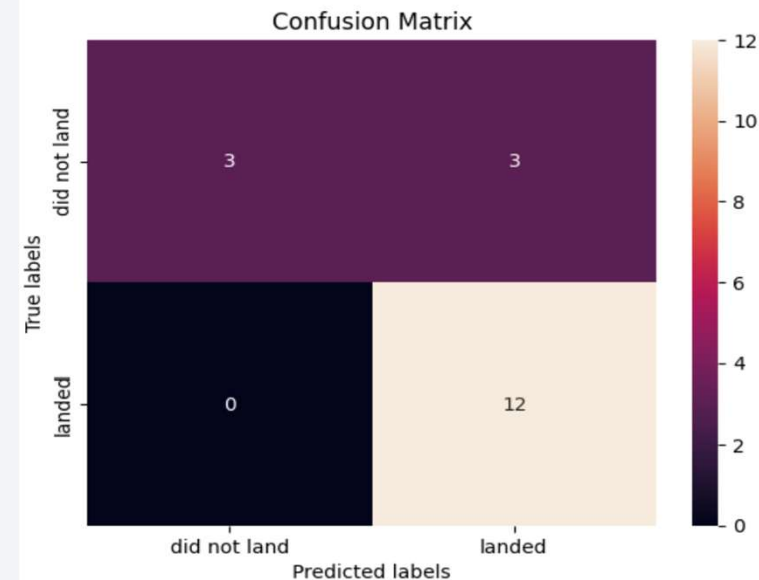
- Model's accuracy score for testing data

```
logreg_cv.score(X_test, Y_test)
```

```
0.8333333333333334
```

Used function to display model's Confusion matrix

```
yhat = logreg_cv.predict(X_test)
plot_confusion_matrix(Y_test, yhat)
```



Appendix

2. Support Vector Machine (SVM) Model

- Created model

```
parameters = {'kernel':('linear', 'rbf','poly','rbf', 'sigmoid'),  
              'C': np.logspace(-3, 3, 5),  
              'gamma':np.logspace(-3, 3, 5)}  
svm = SVC()
```

```
svm_cv = GridSearchCV(estimator=svm, cv=10, param_grid=parameters,verbose=2)  
svm_cv.fit(X_train, Y_train)
```

Fitting 10 folds for each of 125 candidates, totalling 1250 fits

- Tuned model & printed training accuracy & best parameters

```
print("tuned hpyerparameters :(best parameters) ",svm_cv.best_params_)  
print("accuracy :",svm_cv.best_score_)
```

```
tuned hpyerparameters :(best parameters) {'C': 1.0, 'gamma': 0.03162277660168379, 'kernel': 'sigmoid'}  
accuracy : 0.8482142857142856
```

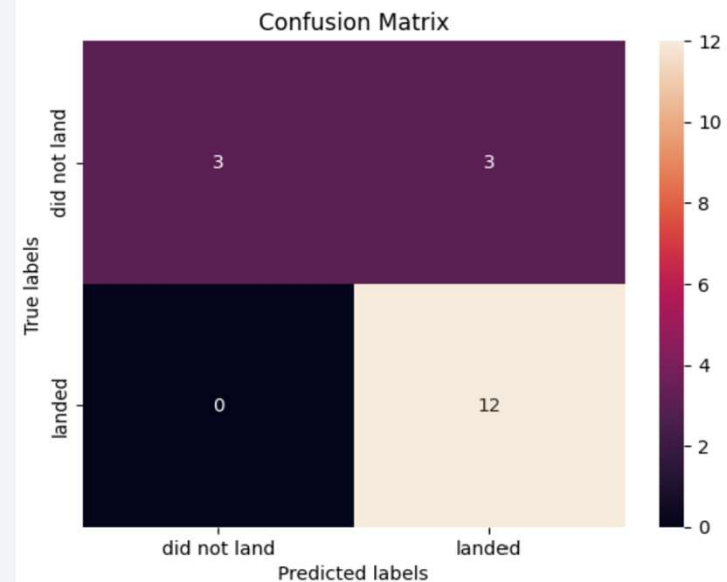
- Model's accuracy score for testing data

```
svm_cv.score(X_test, Y_test)
```

```
0.8333333333333334
```

Used function to display model's
Confusion matrix

```
yhat=svm_cv.predict(X_test)  
plot_confusion_matrix(Y_test,yhat)
```



Appendix

3. Decision Tree Classifier Model

- Created model

```
parameters = {'criterion': ['gini', 'entropy'],
              'splitter': ['best', 'random'],
              'max_depth': [2*n for n in range(1,10)],
              'max_features': ['auto', 'sqrt'],
              'min_samples_leaf': [1, 2, 4],
              'min_samples_split': [2, 5, 10]}

tree = DecisionTreeClassifier()
```

```
tree_cv = GridSearchCV(estimator=tree, cv=10, param_grid=parameters, verbose=1)
tree_cv.fit(X_train, Y_train)
```

Fitting 10 folds for each of 648 candidates. totalling 6480 fits

- Tuned model & printed training accuracy & best parameters

```
print("tuned hyperparameters :(best parameters) ", tree_cv.best_params_)
print("accuracy :", tree_cv.best_score_)
```

```
tuned hyperparameters :(best parameters) {'criterion': 'gini', 'max_depth': 16, 'max_features': 'sqrt', 'min_samples_leaf': 1, 'min_samples_split': 10, 'splitter': 'random'}
accuracy : 0.8767857142857143
```

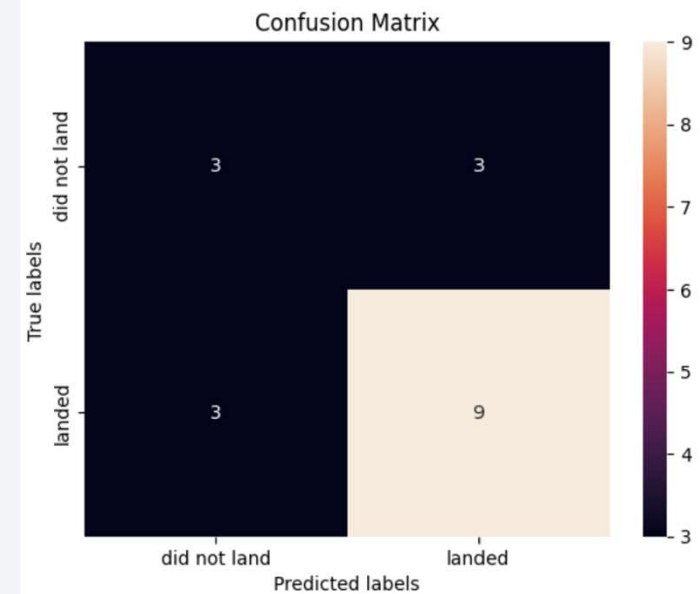
- Model's accuracy score for testing data

```
tree_cv.score(X_test, Y_test)
```

```
0.6666666666666666
```

Used function to display model's Confusion matrix

```
yhat = tree_cv.predict(X_test)
plot_confusion_matrix(Y_test, yhat)
```



Appendix

4. K Nearest Neighbors (KNN) Model

- Created model

```
parameters = {'n_neighbors': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
              'algorithm': ['auto', 'ball_tree', 'kd_tree', 'brute'],
              'p': [1, 2]}

KNN = KNeighborsClassifier()
```

```
knn_cv = GridSearchCV(estimator=KNN, cv=10, param_grid=parameters, verbose=1)
knn_cv.fit(X_train, Y_train)
```

Fitting 10 folds for each of 80 candidates, totalling 800 fits

- Tuned model & printed training accuracy & best parameters

```
print("tuned hpyerparameters :(best parameters) ",knn_cv.best_params_)
print("accuracy :",knn_cv.best_score_)
```

```
tuned hpyerparameters :(best parameters) {'algorithm': 'auto', 'n_neighbors': 10, 'p': 1}
accuracy : 0.8482142857142858
```

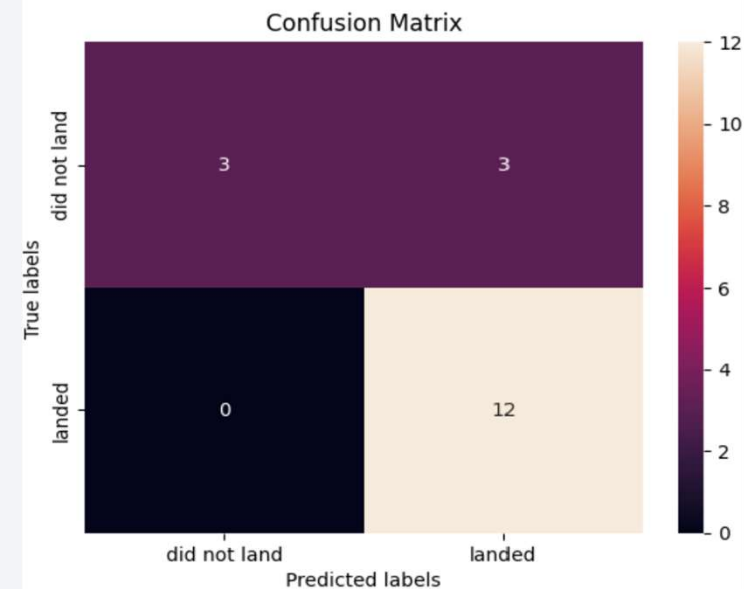
- Model's accuracy score for testing data

```
knn_cv.score(X_test, Y_test)
```

```
0.8333333333333334
```

Used function to display model's Confusion matrix

```
yhat = knn_cv.predict(X_test)
plot_confusion_matrix(Y_test,yhat)
```



Appendix

Find the method performs best:

```
|: results = {  
    "Logistic Regression": logreg_cv.best_score_,  
    "Decision Tree": tree_cv.best_score_,  
    "SVM": svm_cv.best_score_,  
    "KNN": knn_cv.best_score_  
}  
  
best_model = max(results, key=results.get)  
  
print("Best performing model:", best_model)  
print("Accuracy:", results[best_model])
```

```
Best performing model: Decision Tree  
Accuracy: 0.8767857142857143
```

```
|: for model, score in results.items():  
    print(f"{model}: {score}")
```

```
Logistic Regression: 0.8464285714285713  
Decision Tree: 0.8767857142857143  
SVM: 0.8482142857142856  
KNN: 0.8482142857142858
```

References

- Microsoft Bing. (2026). Copilot (September 12 version) [Large Language Model].
<https://copilot.microsoft.com/chats/MQab3W4zwzUrT7JPgRVqW>

Thank you!

